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Parametric modeling and deep learning-based forward and inverse design for acoustic metamaterial plates

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ABSTRACT

Acoustic metamaterials (AMMs), with extraordinary physical properties, could be used to suppress the specific frequencies elastic waves by altering the structure artificially. However, as the complexity of AMMs structures continues to rise, classical design methods are time-consuming and high-computational. Therefore, there is a pressing need to explore more efficient and accurate design methods for AMMs. In this work, a deep learning algorithm-based forward and inverse design method for acoustic metamaterial plates (AMPs) is proposed. First, the initial samples of AMPs are created with parametric model and the bandgaps properties of the AMPs are generated by the finite element method. The dataset consists of different structure parameters and corresponding bandgap characteristics. Then, A neural network model is constructed by concatenating a pretraining network and an inverse design network. Through inputting the dataset to the concatenated network, the mapping relationship between the structural parameters and the bandgap characteristics of the AMPs can be explored. Ultimately, the trained network enables both forward designs, yielding bandgap characteristics for given structural parameters, and inverse design, deducing structural parameters for specific bandgap characteristics. The accuracy of the proposed design methodology is verified through illustrative examples. The results demonstrate that the trained neuron networks can effectively replace the complex physical mechanisms between the structural parameters and bandgap characteristics.

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Acoustic metamaterial;
material design; bandgap;
deep learning; neural
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analysis

1. Introduction

Metamaterial [1, 2] has been a novel research hotspot in the past decades. The formal proposal and validation of elastic wave bandgaps [3–6] indicated the research of acoustic metamaterials (AMMs) got on the right track. The formation of elastic wave bandgaps primarily relies on mechanisms such as Bragg scattering mechanism [7, 8] and local resonance mechanism [9–11]. According to the different acoustic characteristics, AMMs can be applied in fields such as sound focusing, vibration isolation, noise reduction, and acoustic invisibility [12–14]. The acoustic metamaterial plates (AMPs) structures [15–18] are easy to manufacture and has excellent acoustic characteristics and light weight [19–21]. Because of these advantages, AMPs can be usually used as sound insulation and noise reduction interlayer. AMPs has already become a research focus recently.

Existing research on AMPs has primarily focused on analyzing the mechanisms underlying specific structures with unique properties [22–24]. These researches usually propose a new metamaterial structure and verify the properties of the structure theoretically and experimentally [25, 26], while the research on obtaining metamaterial structures based on desired properties is limited. With the advancement of

metamaterials technology, the structures are getting progressively complex, containing more parameters related with the material properties. It is becoming more difficult to use analytical methods to uncover the implicit relationships between parameters and properties. Hence, it is necessary to propose new methods to analyze and study the characteristics of metamaterials. The powerful data analysis capabilities of machine learning methods provide a new option for the design of metamaterials.

Earlier machine learning methods include the perceptron [27], genetic programming [28], and Monte Carlo methods [29]. McCulloch and Pitts [30] introduced the mathematical framework of artificial neuron networks and Ivakhnenko [31] started developing deep learning (DL) models so that the neuron network can solve nonlinear problem. With the assistance of algorithms like backpropagation algorithm [32, 33], DL models showed excellent performance for helping handle nonlinear problems and uncovering complex mapping relationships [34, 35]. Currently, the intelligent methods have been greatly developed and begun to be applied in various engineering fields [36–38].

Some intelligent methods such as genetic algorithms [39], linear regression [40], and Bayesian algorithms [41] have been already used to uncover the complex relationships of

metamaterials [42–45]. In the existing research, DL methods are usually used as auxiliary tools in design work [46, 47]. Especially in the analysis of mechanical properties of materials, DL methods provide researchers with a lot of help. Koya et al. [48] compared different machine learning algorithms in predict mechanical properties of concrete. Mahesh et al. [49] used DL method to analysis coupled frequency of skewed multiphase magnetoelectric composite plates. Nguyen et al. [50] developed a neural network-based fast analysis and prediction approach for geometrically nonlinear bending analysis of plates and shells. Ho et al. [51] developed a DL-based artificial intelligence model for the prediction of Young's modulus of polymer/carbon-nanotubes composites. Liao et al. [52] developed a DL-based inverse design framework for chiral structures and successfully predicted the structural parameters of chiral metamaterials using the framework. Kollmann et al. [53] designed a DL model based on convolutional neural networks for real-time topology optimization of metamaterials with different desired properties. It can be seen that DL methods are widely applied in mechanical properties analysis, but there are few studies on AMPs and acoustic properties.

DL methods have also demonstrated certain potential in the study of bandgap characteristics. Ziletti et al. [54] proposed a DL-based method for automatic classification of crystal structures based on symmetry properties. Chen et al. [55] used multi-objective optimization algorithm to optimize the bandgap of chiral phononic crystals. DL methods can be used for bandgap optimization and classification, but it remains a challenge to design AMPs according to required bandgap range.

In the above studies, the combination of DL and various materials has already helped researchers achieve many good results. However, the research on the design of AMPs based on DL methods remains relatively scarce. The powerful learning ability of DL in dealing with complex mapping relationships may bring a new approach for AMPs design.

In this study, a DL method is applied in AMPs and the mapping relationship between structural parameters and bandgap characteristics is established through the concatenated DNN. The trained concatenated DNN ultimately achieves both forward and inverse design of AMPs. The rest of the paper is organized as follows. Section 2 shows the working principle of AMPs and the method of obtaining AMPs sample set through parametric modeling. Section 3 describes the DL model applied to AMPs and the composition of DNN. Section 4 verifies the accuracy of the output results from the concatenated DNN. Finally, the conclusions are summarized in Section 5.

2. Parametric modeling and sample set of AMPs

2.1. AMPs modeling

The AMPs in this study is composed of basic units arranged periodically. The basic unit contains a square mass frame and a local resonance unit, as shown in Figure 1. When the AMPs are excited by the frequency of external noise source, the local resonance unit will vibrate at the same time. If the

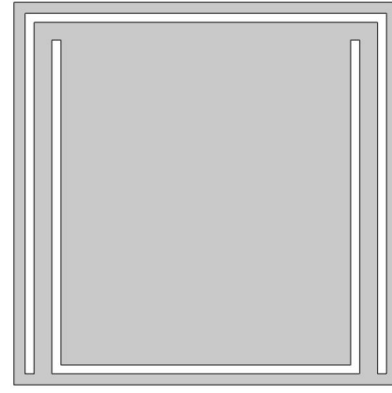


Figure 1. Basic unit of AMPs.

frequency of elastic wave is close to the resonance frequency of the resonance unit, resonance unit and elastic wave will create coupling effect and generate bandgap. Due to the effect of local resonance unit, the AMPs can collect the vibration energy from noise source in the bandgap frequency range to achieve noise reduction.

2.2. Bandgap characteristics of AMPs

Since the AMPs is symmetric and composed of periodic arrangement of square elements, according to Bloch theorem [56], the eigen field of a linear system with translational periodicity can be expressed as Eq. (1).

$$u(r) = U_k(r)e^{i(k \cdot r)} \quad (1)$$

where k is Bloch wave vector, r is position vector, and $u(r)$ is amplitude function.

The plane wave vector denotes the phase change of the wave function between different cells, and its magnitude corresponds to the wave function only in a reciprocal lattice. The propagation behavior of elastic waves in the entire AMPs can be converted to a single unit and its first irreducible Brillouin region [57]. The Brillouin region of the square unit is shown in Figure 2(a), where M- Γ -X-M is the first irreducible Brillouin region of the square unit.

To satisfy Bloch theorem for every point in the square element, the boundary displacement of the constraining structure is set using Floquet periodic boundary, which satisfies the equation Eq. (2).

$$u_t = u_s \exp[-ik_F \cdot (r_t - r_s)] \quad (2)$$

where u_t and u_s denote the target end and source end of Floquet periodic boundary conditions, respectively.

The wave vector k is made to traverse once along the boundary of the first irreducible Brillouin region M- Γ -X-M. When defining the wave vector k , two edges or boundaries in the x and y directions are taken as two source terms, and a periodic boundary condition is applied in the x and y directions of the basic element.

After the range is determined, parametric scanning was set for the wave vector k . The scanning range was 0–3 and the step length was 0.01. The eigenfrequency solver is used to solve the eigenvalue equation (Eq. 3) to obtain the

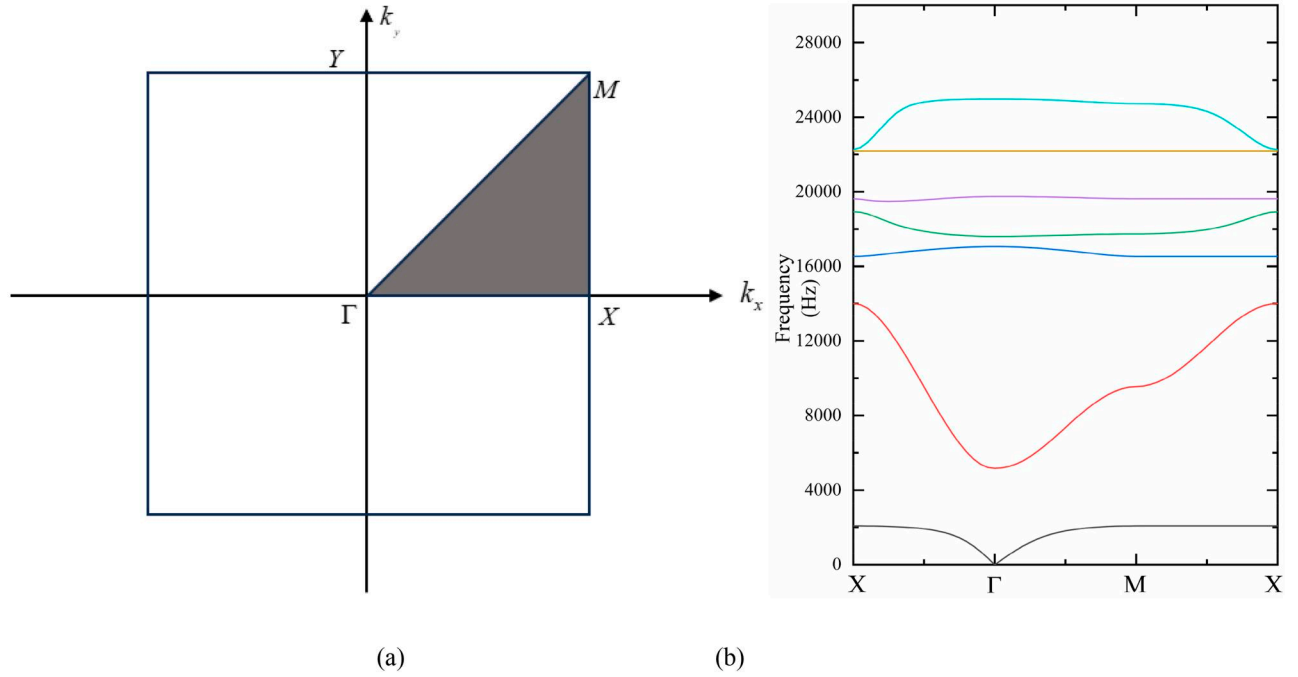


Figure 2. (a) Brillouin region of the basic unit and (b) band diagram of basic unit.

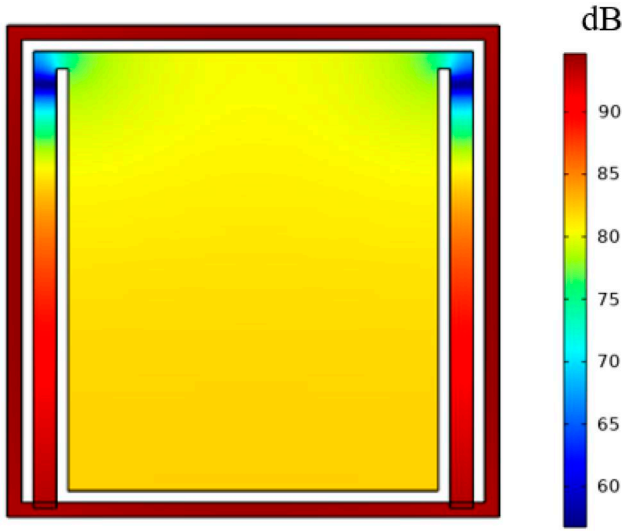


Figure 3. Sound pressure level distribution diagram of basic unit at characteristic frequency of 5157 Hz.

characteristic frequencies under different k and the complete band map, as shown in Figure 2(b).

The eigenvalue equation can be written as

$$(K - \omega^2 M)u(\nu) = F \quad (3)$$

where K denotes element stiffness matrix; M denotes mass matrix; $u(\nu)$ denotes the displacement of the generalized node; and F denotes the vector of force.

According to the band diagram, the basic AMPs structure has a bandgap with the range of 2031–5157 Hz. The sound pressure level distribution diagram of basic unit is shown in Figure 3. The sound pressure level in square frame is much higher than that in local resonance unit at the frequency of 5157 Hz, which suggests that most acoustic energy is concentrated in square frame.

2.3. Dataset of AMPs with different structural parameters

DL requires a huge number of datasets to support the training of DNN so that the DNN can obtain the desired result. A dataset containing huge number of AMPs samples is needed. Without changing the side length L of the square mass frame, four structural parameters affect each basic unit, which are the width of the square mass frame, the distance between the square mass frame and the cantilever beam, the width of the cantilever beam and the distance between the cantilever and the internal rectangular plate. As depicted in the Figure 4, the four structural parameters are defined as variables a , b , c , and d , respectively, and some samples with different structural parameters are plotted in Figure 5.

The range of four variables is set as 2–4.5 mm, and the step size was set as 0.5. Each variable has six different values. A total of 1296 different models is obtained by using MATLAB R2021a and COMSOL Multiphysics® 6.0 co-simulation. The upper and lower bounds of the bandgaps in the band diagram of each model are extracted, and the obtained 1296 sets of models are taken as the samples of DL.

3. The framework of deep learning models for AMPs

In this section, the framework of the deep learning models adopted in the design of AMPs is elaborated. The current mainstream DL frameworks mainly include TensorFlow and PyTorch. Due to the excellent scalability and compatibility with other coding languages, TensorFlow is used as the software environment in this study.

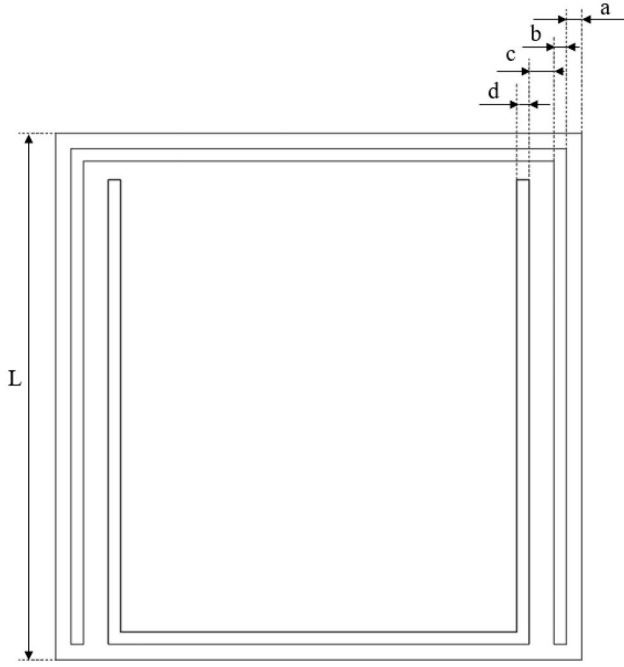


Figure 4. Main structural parameters of basic unit.

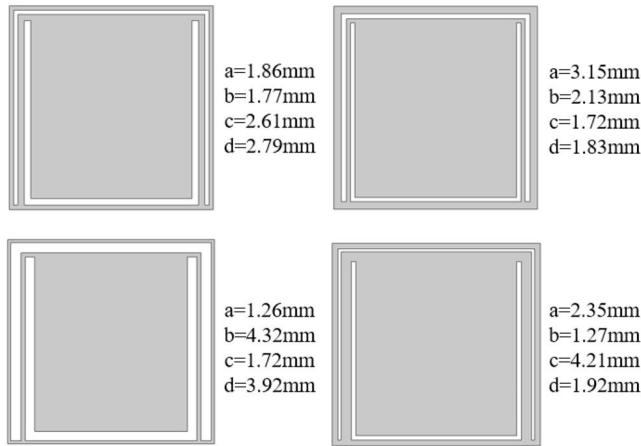


Figure 5. Basic units consisting of different structural parameters.

3.1. Forward pretraining network

A concatenated DNN structure is adopted to solve the problem. The concatenated DNN includes a pretraining network and an inverse design network, as shown in Figure 6.

The pretraining network is a feed-forward network used to train the weights and biases from the structural parameters to bandgap characteristics of AMPs. The trained weight and biases are then used to assist in the training of the inverse design network. The pretraining network is built upon a backpropagation neural network which adjusts the weights and biases of neurons in each layer based on error signals. The structural parameters a , b , c , and d are used as the input layer of the pretrained neural network, and the bandgap range data is used as the output layer of the neural network, which includes three hidden layers with 100, 100, and 50 nodes, respectively.

During forward propagation, the output of each layer serves as the input to the next layer, and the relationship between the weight matrix and bias matrix of each hidden layer is shown in Eq. (4).

$$y = w_i x + b_i \quad (4)$$

where y denotes the output of each layer, w_i denotes the weight matrix of the i th hidden layer, and b_i the bias matrix of the i th layer. The initial weight matrix and bias matrix of each layer are defined as truncated normal distribution random matrices. The random numbers generated in the matrix should not exceed two times the standard deviation from the mean so as to avoid significantly different results in each training.

Sigmoid activation function (Eq. 5) is used in each hidden layer to restrict the output to the range of 0 to 1 so that the model can handle nonlinear problems, as shown in Figure 7(a). The predicted structural parameters are also restricted within the range of the simulated features in the training set by feature normalization. In the backpropagation of the error, sigmoid function's derivative (Figure 7(b)) is used to adjust the weight and bias.

$$S(y) = \frac{1}{1 + e^{-y}} \quad (5)$$

The loss function can quantify the difference between the output value and the target value. The mean squared error

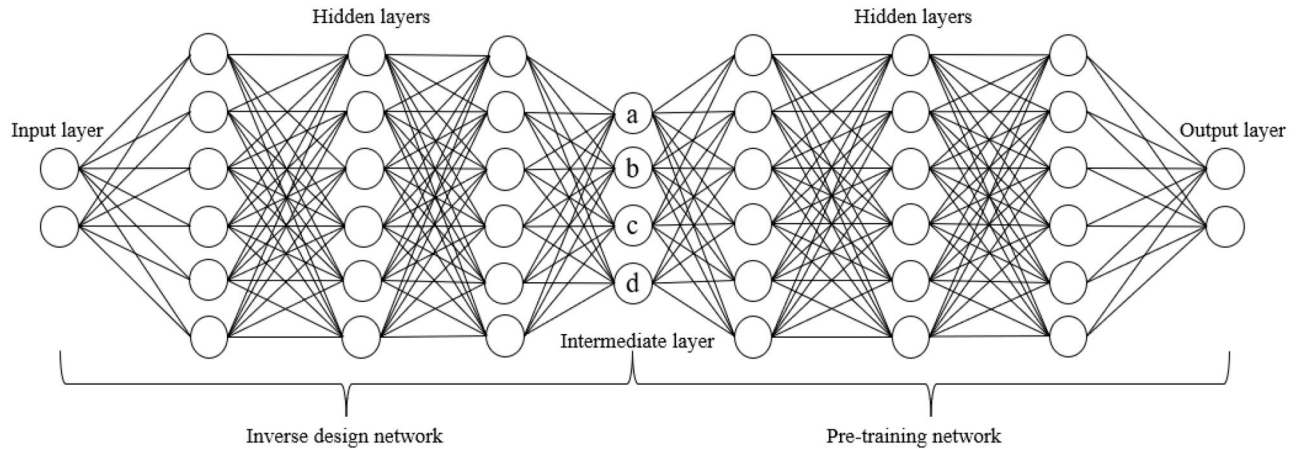


Figure 6. The structure of concatenated DNN.

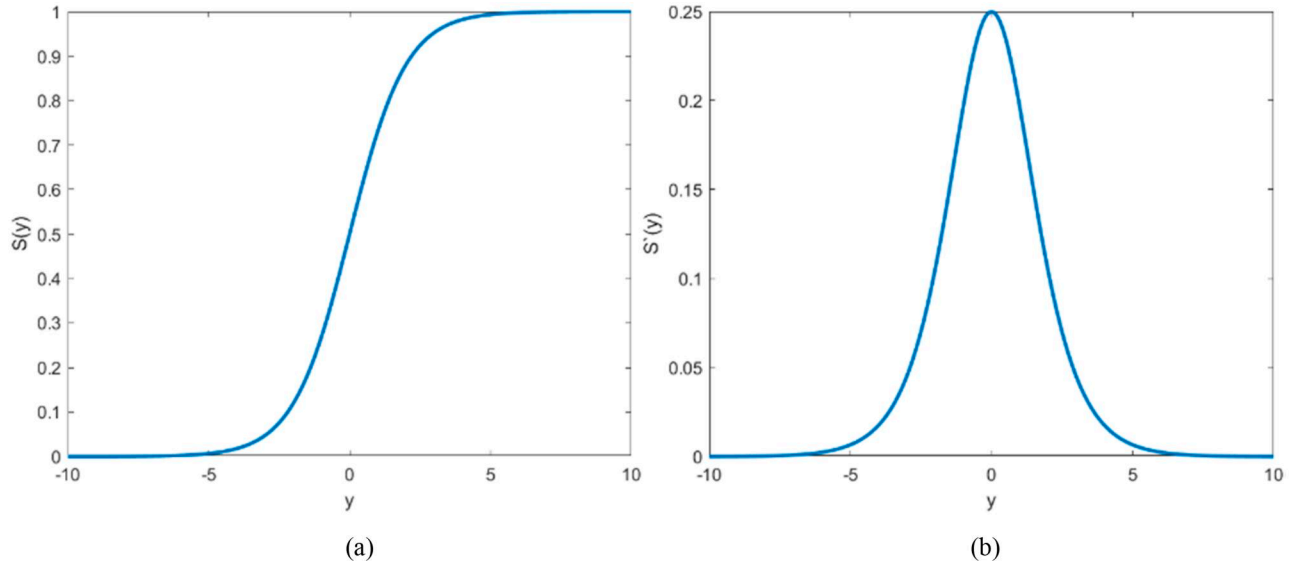


Figure 7. (a) Image of Sigmoid function and (b) image of sigmoid function's derivative.

loss function (Eq. 6) is used in the pretraining network.

$$\text{MSE}(y, y') = \frac{\sum_{i=1}^n (y_i - y'_i)^2}{n} \quad (6)$$

where y_i is the true value of the i th data in the output, and y'_i is the predicted value given by the neuron networks, which denotes the true and predicted ranges of bandgaps.

The optimized Adaptive Moment Estimation (Adam) algorithm [58] (Eq. 7) is used in the pretraining network to minimize the loss function. Adam algorithm can adjust the learning rate of each parameter dynamically by using the first and second moment estimates of the gradient.

$$lr_t = lr * \sqrt{1 - \text{beta}_2^t} / (1 - \text{beta}_1^t) \quad (7)$$

where lr is the learning rate parameter, beta_1^t is the exponential decay rate for the first moment estimates, and beta_2^t is the exponential decay rate for the second moment estimates.

3.2. Inverse design network

To overcome the non-uniqueness issue in inverse design, the pretraining network is used to assist the inverse design network training so that the concatenated DNN can be trained on samples without matching the actual conditions. The concatenated DNN only needs to produce results similar to the actual ones, and the loss function will decrease. The loss function is defined as the mean squared difference between the desired bandgap range and the predicted bandgap. The target bandgap range is the input of inverse design network and the structural parameters of the AMPs are outputs. Conversely, the pretraining network takes as input the structural parameters generated by the inverse design network. The neurons in the intermediate layers connecting the two networks represent the structural parameters a , b , c , and d , as shown in Figure 9. The trained weights and biases of the pretraining network are used to predict the corresponding bandgap range from the structural

parameters output by the inverse design network. The concatenated DNN also utilizes backpropagation neuron networks and the Adam optimizer for training to minimize the loss between the desired bandgap range and the model-predicted bandgap.

The detailed workflow is shown as Figure 8. After the training of the concatenated DNN is completed, the inverse design network can be separated from the concatenated DNN and used as an independent model to output the structural parameters of the AMPs based on the desired objectives.

4. Forward and inverse design results through the concatenated DNN

4.1. Forward design of bandgap range based on structural parameters

In the pretraining network, 80% of the samples were divided into a training set, and the remaining 20% were used as a test set. After iterative training with the pretraining network, the difference between the output values and the target values was quantified by the loss function to reflect the training status of the DNN. The convergent loss function curve can prove that the DNN has been effectively trained. The training process and all numerical experiments were performed on a laptop computer with a GPU of NVIDIA GeForce GTX 3060, a CPU of AMD core Ryzen 9 5900HX. The trained loss curve for the first 300 iterations of the pretraining NN is shown in the Figure 9. Both the training and testing loss functions have converged, indicating that the pretraining network has been well trained. The total time for 300 iterations of the pretraining network is 39.73 s, and the average time for each iteration is 0.13 s.

The pretraining neuron network model can be used for the forward design of AMPs, and the bandgap range can be quickly obtained by inputting structural parameters. Randomly inputting six sets of structural parameters, we obtain the predicted bandgap ranges using the pretraining

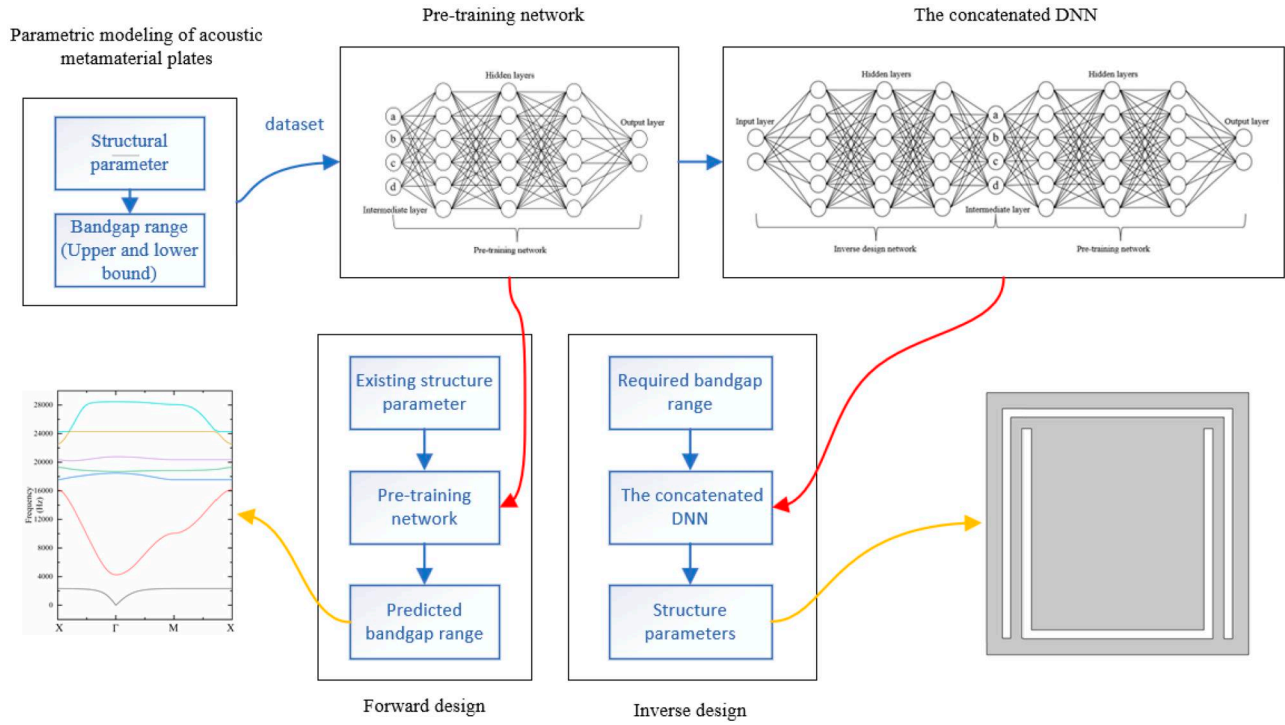


Figure 8. Workflow of AMPs forward and inverse design based on deep learning.

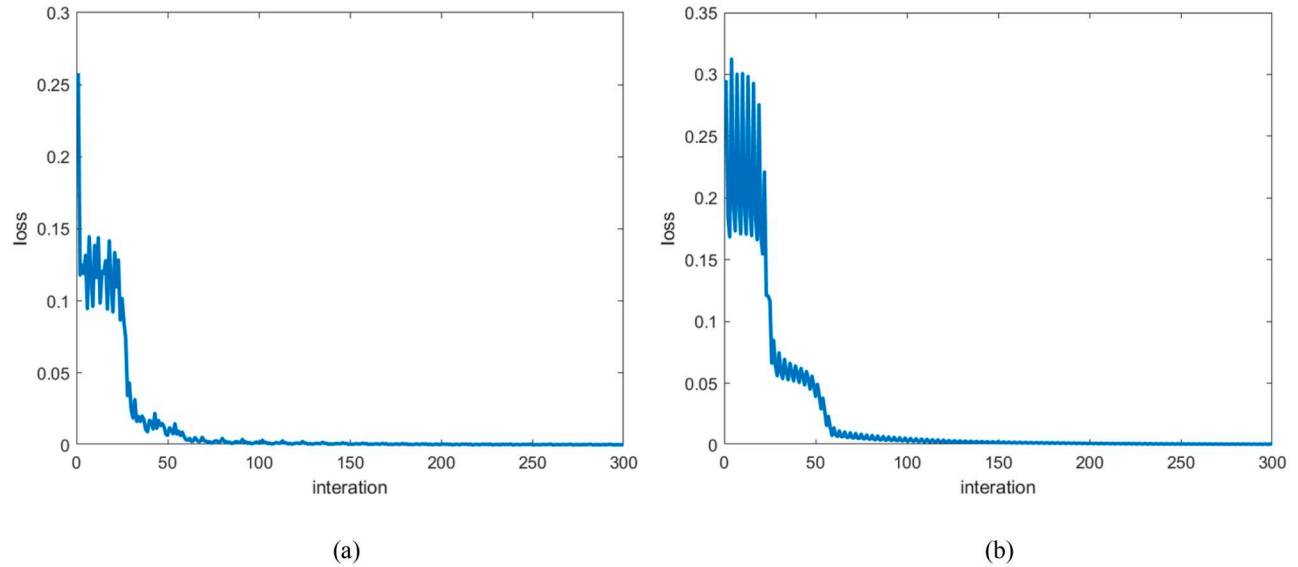


Figure 9. (a) Loss function diagram of training set and (b) loss function diagram of test set.

network and the simulated bandgap ranges using simulation software, respectively. The results are shown in Table 1. The comparison between the predicted bandgap ranges from the pretraining network and the bandgap ranges obtained from simulation is depicted in Figure 10. The two sets of results are nearly identical, indicating that the pretraining network possesses excellent forward design capability.

4.2. Inverse design of structural parameters based on bandgap range

In the concatenated DNN, the loss is defined as the mean squared error between the required and the predicted

Table 1. Comparison between the bandgap range output by the network for randomly input structural parameters and the simulated bandgap range.

Sample	a (mm)	b (mm)	c (mm)	d (mm)	Network output (Hz)	Simulation result (Hz)
1	3.12	1.56	2.33	3.37	3849–1657	3842–1665
2	1.73	3.27	3.32	2.69	5279–1950	5362–1911
3	2.64	2.43	3.62	1.29	4979–1975	4920–1995
4	2.25	3.75	3.75	2.15	5307–2079	5270–2084
5	1.86	1.77	2.61	2.79	4739–1690	4717–1657
6	3.69	2.21	1.51	3.15	3367–1421	3342–1386

bandgap ranges. The optimal model is obtained by minimizing the bandgap characteristics in the input samples and the predicted bandgap characteristics output by the pretraining network during training. The loss function curve is shown

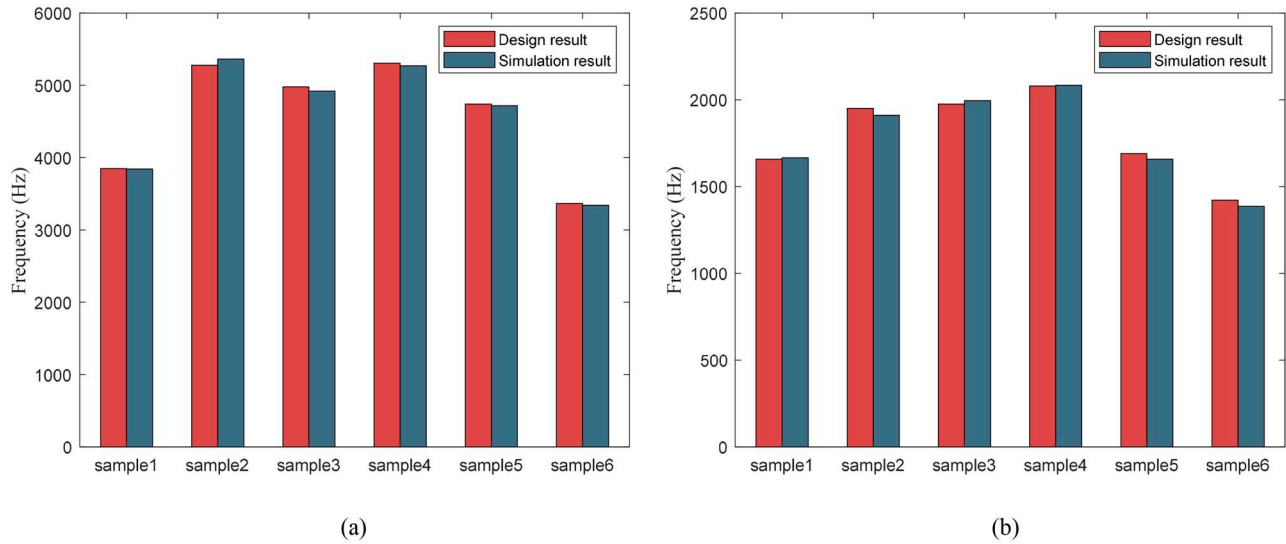


Figure 10. (a) Comparison of designed upper bound and simulation upper bound (b) comparison of designed lower bound and simulation lower bound.

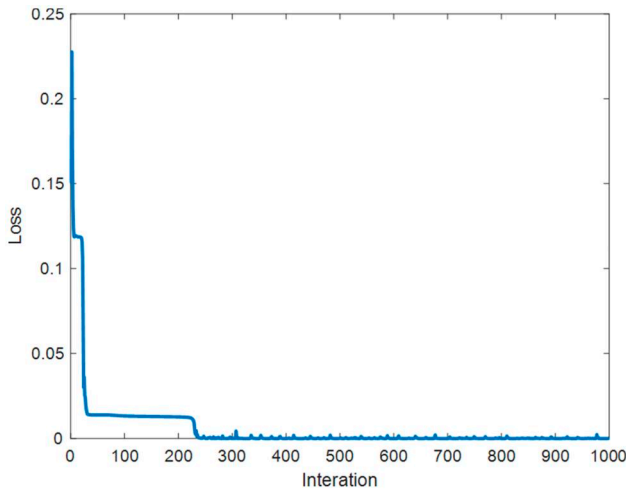


Figure 11. Loss curve of concatenated DNN.

Table 2. The structural parameters of the AMPs obtained by inputting the required bandgap range.

Input bandgap range (Hz)	<i>a</i> (mm)	<i>b</i> (mm)	<i>c</i> (mm)	<i>d</i> (mm)
1500–4500	2.28	3.21	2.56	3.19
1500–5000	1.15	0.72	2.44	4.81
2000–4500	3.74	3.16	3.27	3.31
2000–6000	1.61	3.08	3.95	2.35
2500–5000	3.65	1.97	4.69	3.13
2500–6000	1.58	3.84	4.96	4.25

Table 3. The simulated bandgap range of the AMPs corresponding to the output structural parameters.

<i>a</i> (mm)	<i>b</i> (mm)	<i>c</i> (mm)	<i>d</i> (mm)	Simulation result (Hz)
2.28	3.21	2.56	3.19	1741.1–4461.3
1.15	0.72	2.44	4.81	1569.2–5210.6
3.74	3.16	3.27	3.31	2094.7–4340.9
1.61	3.08	3.95	2.35	2047.4–5815
3.65	1.97	4.69	3.13	2440.3–5045.2
1.58	3.84	4.96	4.25	2389–6355.1

in Figure 11, and the concatenated DNN has been effectively trained. The total time for 1000 iterations of the concatenated DNN is 193.4 s, and the average time for each iteration is 0.19 s.

After the concatenated DNN has been trained, the inverse design network can be extracted to output corresponding structural parameters of the AMPs on the basis of inputting the required bandgap range, as shown in Table 2.

To validate the accuracy of the bandgap characteristics obtained by the inverse design network, the structural parameters output by the concatenated DNN will be used in the AMPs, which will be analyzed using COMSOL software. The simulation results are shown in Table 3, and Figure 12 shows a comparison between the required and the simulated bandgap ranges using the structural parameters output by the inverse design network. The blue part denotes the simulated bandgap range, the orange part denotes the required one, and the purple part denotes the overlap between them. These ranges are basically consistent, indicating the good accuracy of this design method.

4.3. Experimental verification

The experiment measured the sound transmission loss of the sound insulation box with AMPs by measuring the noise passing through it. The sound insulation ability in the bandgap frequency range of AMPs is verified by the acoustic transmission loss experiment. AMPs samples with simulated bandgap ranging from 1569.2 to 5210.6 Hz are prepared, and the output structural parameters are rounded during preparation. The position of the AMPs in the sound insulation box is shown in Figure 13.

A noise source was placed inside the sound insulation box. After determining the position of the sound insulation box, a noise sensor was placed at a distance of one meter in front, behind, left, right, and above the sound insulation box in five directions. The noise level transmitted through the sound insulation box was recorded using the noise sensors, as shown in Figure 14. The sound transmission loss is obtained by subtracting the measurement data of sensors in the different cases of noise passing through the sound insulation box with and without AMPs intermediate layer.

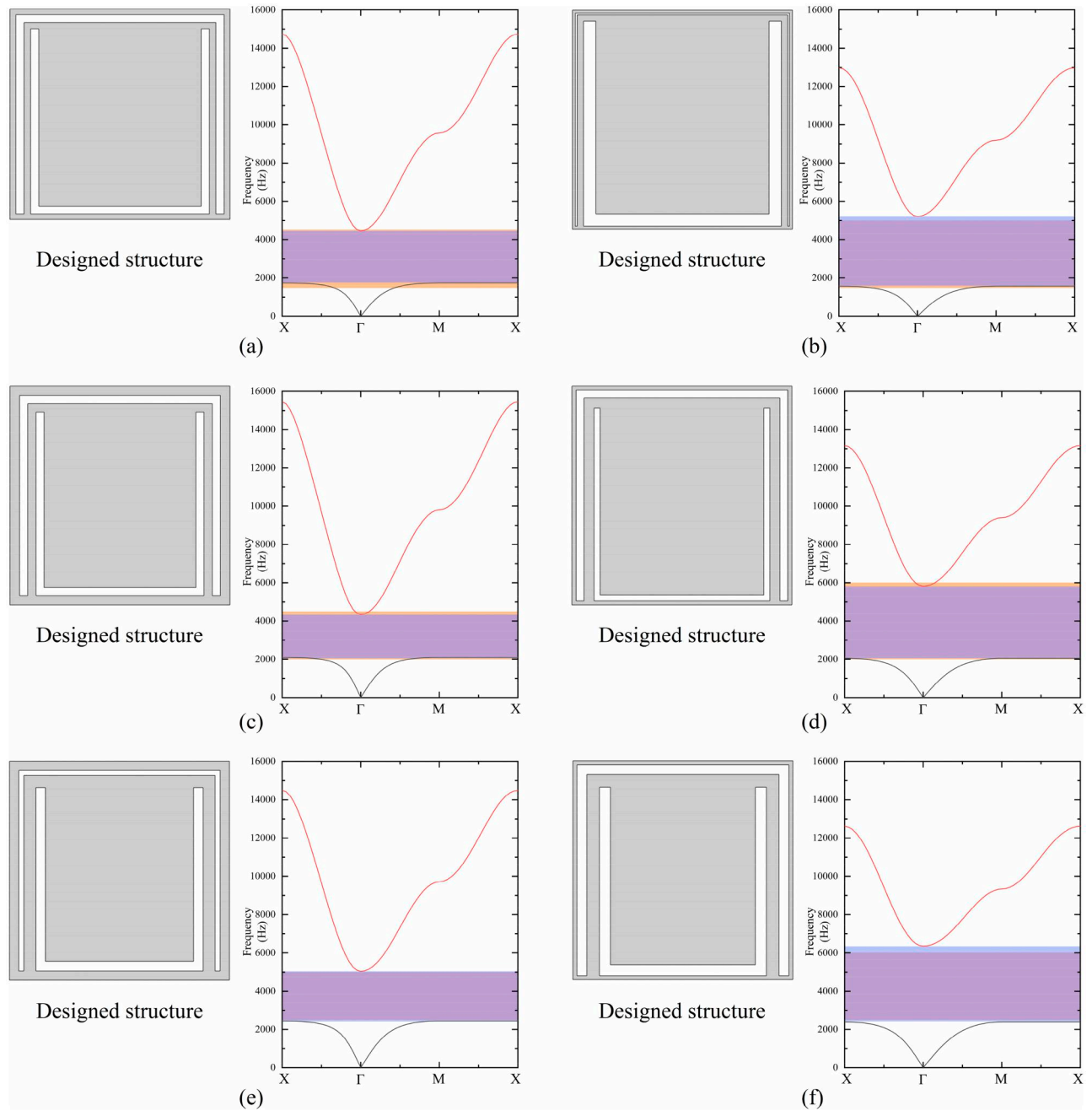


Figure 12. (a–f) basic units and the corresponding bandgap diagrams with different structural parameters.

The sound transmission loss of the AMPs obtained based on the mass law [59] (Eq. 8) is shown as the blue line in Figure 15.

$$\text{STL} = 10\log_{10} \left| 1 + \frac{i\omega m \cos \theta}{2\rho_0 c_0} \right| \quad (8)$$

where STL denotes the sound transmission loss based on the mass law, i denotes the imaginary unit, m denotes the surface mass density, ω denotes the angular frequency, θ denotes the incident angle of the incident sound wave, ρ_0 denotes the air density, and c_0 denotes the speed of sound

in air. Since the incident sound wave is perpendicular to AMPs, the incidence angle θ in Eq. (8) is 0° .

The experimentally measured acoustic transmission loss data of the AMPs interlayer is shown as the orange line in Figure 15. The measured acoustic transmission loss exhibits two peaks compared to the blue line representing mass law curve when subjected to noise in the frequency range of 1569.2–5210.6 Hz. The comparison can indicate that the local resonant structures of the AMPs effectively block sound energy and align with the obtained bandgap characteristics. The experimental results show that the bandgap characteristics obtained by inverse design are accurate.

5. Conclusion

This study proposed a new design method by combining DL with the design of AMPs. A pretraining network and an inverse design network are concatenated to explore the non-linear mapping relationship between the structural parameters of the AMPs and bandgap characteristics. Conclusion remarks are drawn as follows:

1. The trained concatenated DNN model enables both forward design and inverse design of the AMPs. The accuracy of the concatenated DNN model is validated by comparing multiple sets of output results with simulation results. The concatenated DNN can accurately

express the mapping relationship between the structural parameters and bandgap characteristics of AMPs, and can effectively replace complex physical mechanisms.

2. The training time of concatenated DNN model is short and the trained concatenated DNN model can directly output the desired structural parameters or bandgap range. The purpose of design according to the requirements can be achieved without using extensive simulation and experimental verification to seek suitable structural parameters.
3. This method exhibits good scalability as it can be applied to a wider range of research fields by modifying the input and output of the DNN.

This work only focuses on the relationship between structural parameters and bandgap characteristics in AMPs. In future works, we will try to advance the research on meta-material properties by leveraging the unique learning

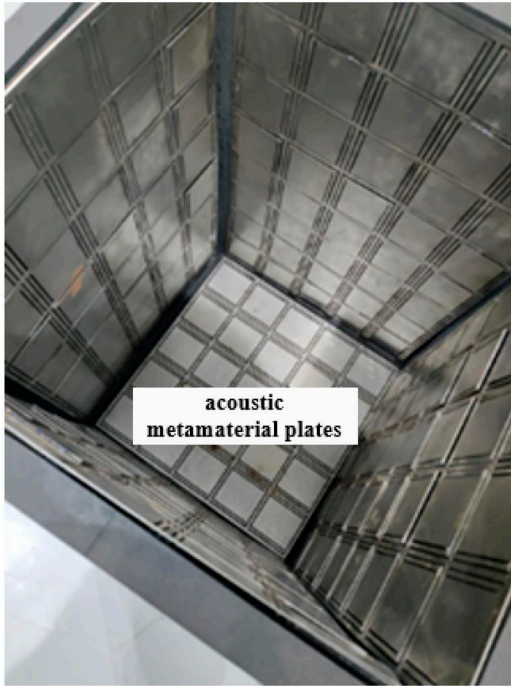


Figure 13. Arrangement inside the sound insulation box.

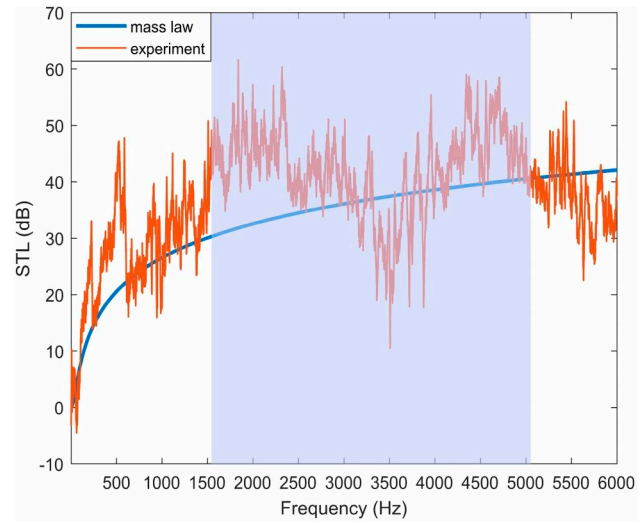


Figure 15. Experimental sound transmission loss and sound transmission loss under mass law.

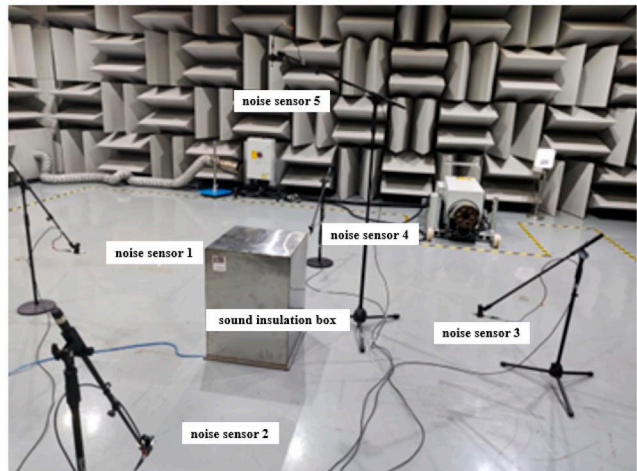
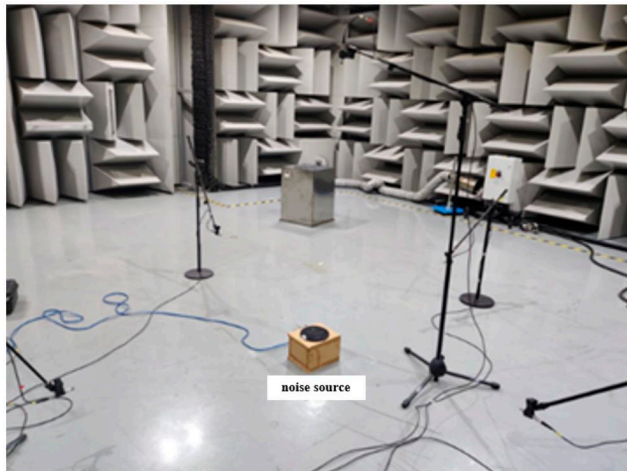


Figure 14. Arrangement of the noise source and noise sensors.

mechanisms and nonlinear representation capabilities offered by DL. Additionally, not limited to DL, the combination of various other intelligent algorithms will also provide researchers with more efficient and accurate design methods.

Authors' contributions

Weiqian Chen: Visualization, Conceptualization, Validation, Investigation, Methodology, Writing - Original Draft, Data Curation. **Hui Guo:** Formal analysis, Project administration, Supervision, Methodology, Investigation, Writing - Review & Editing, Funding acquisition, Resources. **Yansong Wang:** Supervision, Funding acquisition, Resources. **Fuyin Ma:** Supervision, Methodology, Software. **Pei Sun:** Project administration, Supervision, Investigation, Methodology, Writing - Review & Editing, Funding acquisition, Resources, Software. **Tao Yuan:** Supervision, Methodology, Software. **Xiaolong Xie:** Supervision, Methodology, Software.

Disclosure statement

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